

① Three cells each of emf E but internal resistance $2r, 3r$ and $6r$ are connected in parallel across a resistor R .

Obtain expressions for (i) current flowing in the circuit and (ii) the terminal potential difference across the equivalent circuit cell.

② Explain how free electrons in a metal at constant temperature attain an average velocity under the action of electric field. Hence obtain an expression of it.

③ Using Huygen's construction, show how a plane wave is reflected from a surface. Hence, verify the laws of reflection.

④ An alternating voltage of $220V$ is applied across a device X . A current of $0.22A$ flows in the circuit and it lags behind the applied voltage in phase by $\frac{\pi}{2}$ radian. When the same voltage is applied across another device Y , the current in the circuit remains the same, and it is in phase with the applied voltage.

(i) Name the devices X and Y .

(ii) Calculate the current flowing in the circuit when the same voltage is applied across the series combination of X and Y .

⑤ Which of the following has permeability less than that of free space?

A) Copper B) Aluminium C) Copper chloride D) Nickel